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Enhanced consensus control architecture for autonomous platoon utilizing multi-agent reinforcement learning

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Abstract

Coordinating a platoon of connected and automated vehicles significantly improves traffic efficiency and safety. Current platoon control methods prioritize consistency and convergence performance but overlook the inherent interdependence between the platoon and the the non-connected leading vehicle. This oversight constrains the platoon's adaptability in car-following scenarios, resulting in suboptimal optimization performance. To address this issue, this paper proposed a platoon control framework based on multi-agent reinforcement learning, aiming to integrate cooperative optimization with platoon tracking behavior and internal coordination strategies. This strategy employs a bidirectional cooperative optimization mechanism to effectively decouple the platoon's tracking behavior from its internal coordination control, and then recouple it in a multi-objective optimized manner. Additionally, it leverages long short-term memory networks to accurately capture and manage the platoon's dynamic nature over time, aiming to achieve enhanced optimization outcomes. The simulation results demonstrate that the proposed method effectively improves the platoon's cooperative effect and car-following adaptability. Compared to the consensus control strategy, it reduces the average spacing error by 8.3%. Furthermore, the average length of the platoon decreases by 19.1%.

1 | INTRODUCTION

The theoretical underpinnings established by intelligent transportation systems (ITS) research have provided indispensable foundations for the evolution of connected and autonomous vehicles (CAVs) (Adeli & Ghosh-Dastidar, 2004; Hooshdar & Adeli, 2004). Through the development of vehicle-to-everything communication architectures and multi-agent coordination frameworks, ITS advancements enable CAVs to achieve superior environmental perception capabilities, compared to conventional autonomous systems (Lesch et al., 2022). Central to the functionality of CAVs is platooning, which promotes cooperation and coordination between vehicles by facilitating information sharing. To achieve effective platoon control, it is essential for following vehicles to react quickly to the movements of the platoon leader while maintaining a safe distance and consistent speed (Lei et al., 2015). Notably, platoon driving can achieve up to 16% energy savings (T. Liu et al., 2023). It can also improve traffic flow and increase the capacity of existing road infrastructure by up to 200%, reducing congestion (Y. Liu et al., 2023).

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Existing studies on vehicular platoon control predominantly classify control methodologies into two categories: model-based approaches and learning-based approaches. Model-based approaches typically employ nonlinear control strategies to characterize vehicle dynamics and capture inter-vehicle interactions within platoons. Representative techniques include integral sliding-mode control (ISMC; Y. Li et al., 2019), sliding-mode control (Wang et al., 2020), cooperative adaptive control (Ruan et al., 2025), and $H\infty$ control (Yang Zheng et al., 2018). The core challenge in platoon control lies in making real-time decisions within dynamic systems amidst various uncertainties. To address these challenges, several studies have explored the application of model predictive control (MPC; Dong et al., 2025; Pauca et al., 2023) and consensus control strategies (CCS). For example, a longitudinal controller based on MPC has been developed for string stability analysis (Pauca et al., 2023). In order to address the online computation and control challenges, the literature (Zhang & Du, 2025) proposes the use of a real-time executable hierarchical and recursive platoon formation architecture for the platoon control. In addition, previous study (Chen et al., 2021) proposed a CCS for vehicle platoons and theoretically investigated the asymptotic stability using the Lyapunov theorem. The reference (Lyu et al., 2025) proposes a data-driven robust predictive control method, which has been demonstrated to effectively solve traffic oscillations and improve traffic efficiency. Their results demonstrate the effectiveness of the proposed approach in achieving platoon convergence and maintaining consistent control. Despite the advantages of model-based control methods in improving platoon stability, developing accurate and reliable models for the longitudinal motion of vehicle platoons remains a significant challenge owing to the complex nature of nonlinear dynamics. Additionally, model-based control methodologies require manual parameter calibration, a process that is both resource-intensive and time-consuming, while demonstrating limited adaptability in adapting to diverse operational scenarios.

Learning-based methodologies, characterized by their model-agnostic nature, have demonstrated superior performance in ITS control applications (Karim & Adeli, 2003; Rafiei et al., 2024a, 2024b). Current research identifies three dominant learning-based control paradigms: Iterative learning-based control (ILC; Dekker et al., 2019), neural network (NN)-based adaptive control (X. Ma, 2006), and RL-based control (RL). NN and their derived controllers are limited to mimicking the parameter adjustment strategies shown by experts, and their ability to generalize must be validated. While ILC requires minimal prior knowledge, it suffers from computational latency that impedes real-time implementation, posing challenges for real-time control. RL achieves effective control outcomes by learning and enhancing performance through interactions with the environment without the need for precise physical models and has received significant attention in recent years. A comparative analysis (Yuan Lin et al., 2021) was conducted to evaluate the RL and model-based control in terms of cooperative adaptive cruise control (CACC) performance, and the results indicated that the RL algorithm outperformed MPC in uncertain conditions. Previous research (Y. Ma et al., 2021) proposed a longitudinal car-following control algorithm based on a deep deterministic policy gradient (DDPG) RL algorithm. Although platoon-level control ensures individual agent coordination, it fails to account for spatiotemporal interaction dynamics between adjacent vehicles. Focusing on the overall vehicle platoon, a robust control method has been proposed that possesses the capability to adapt to model errors and environmental uncertainties based on RL (Neto & Mozelli, 2024). Despite demonstrating model-free control efficacy, RL-based approaches face fundamental challenges in guaranteeing vehicular platoon string stability.

To overcome these limitations, recent advances focus on hybrid control architectures that synergistically integrate model-based control with RL for enhanced platoon coordination. T. Liu et al. (2023) developed a DDPG-driven state-space compression algorithm combined with dynamic programming (DP), demonstrating improved platoon convergence through dimension reduction. However, by their nature, DP-based strategies are limited to offline computation. In order to address this shortcoming, Yan et al. (2022) proposed a hybrid car-following strategy that integrates DDPG with CACC. In addition, previous study (Yang et al., 2021) proposed a longitudinal control method based on proportional-integral-derivative (PID) and DDPG, the method uses a data-driven approach to achieve real-time automatic adjustment of PID control parameters. Current hybrid control systems predominantly adopt either parallelized architectures or cascaded structures, focusing primarily on intra-platoon coordination. These methodologies seek to achieve consensus on the internal spacing and speed of the vehicles within the platoon. However, the overall consistency of the platoon with the external environment is not addressed, which may result in sub-optimal optimisation performance.

Fundamentally, platoon control can be characterized as a sequential stochastic decision-making problem. Prior research (Chen et al., 2021; Y. Li et al., 2022) has demonstrated that CCS offers a more robust solution for maintaining platoon stability and consistency. Multi-agent RL (MARL) has garnered considerable attention in recent years across various domains, including ground robot control and drone swarm management (Gronauer & Diepold, 2022). MARL has been utilized to address the convergence

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challenges in non-cooperative Markov games and to facilitate collaboration among multiple agents through shared training frameworks. Consequently, numerous studies on platoon control have been conducted based on MARL, yielding effective control outcomes (Chen et al., 2022; M. Li et al., 2025; Shi et al., 2025; Xu et al., 2024). Long short-term memory (LSTM) networks, known for their proficiency in managing sequential data and capturing long-term dependencies, are particularly effective in filtering irrelevant historical information. This makes LSTM highly promising for modeling car-following behaviors (Yangxin Lin et al., 2022). Given these strengths, a control method is proposed that integrates MARL and LSTM network to improve the optimization results. The proposed methodology aims to address the joint optimization of platoon internal consistency control and adaptability in car-following scenarios.

The key contributions of this research are summarized below:

1. To achieve platoon internal consistency control in concert with car-following behaviors, a bidirectional control mechanism is designed. Specifically, by considering the platoon leader as a central hub, a comprehensive trade-off between platoon leader tracking and internal collaborative performance is made to enhance the car-following scenario adaptability of the platoon.
2. A model-based method is deeply integrated with an RL approach, unifying the overall coordinated control of the platoon and the adjustment of CCS parameter weights into a Markov decision process (MDP).
3. To enhance platoon coordination and car-following adaptability, we propose an LSTM based multi-Aagent proximal policy optimization (LSTM-MAPPO) strategy that effectively captures the dynamic interactions between the platoon leader and followers. Specifically, the LSTM's ability to model long-term dependencies enables followers to predict the leader's intentions based on historical data, while the bidirectional control mechanism allows the leader to actively respond to follower feedback, thereby establishing a closed-loop optimization and improving the overall optimization performance.

The subsequent sections of this paper are structured as follows: Section 2 outlines the problem description and system model. Section 3 introduces the LSTM-MAPPO control architecture. Section 4 evaluates the performance of LSTM-MAPPO through simulation and compares it with benchmark algorithms. Finally, Section 5 provides concluding remarks on this paper along with proposed future research directions.

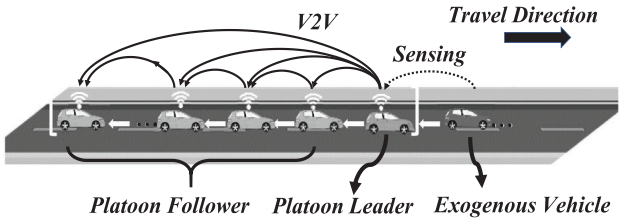


FIGURE 1 Predecessor–leader following topology-based platoon scenario.

2 | SYSTEM MODEL DESCRIPTION

In the scenario illustrated in Figure 1, a platoon of $N+1$ CAVs follows an non-connected vehicle (hereinafter defined as an exogenous vehicle) operating freely on a straight road. The communication between the CAVs is based on the predecessor–leader following topology, and the platoon is assumed to be homogeneous.

2.1 | Platoon model description

Assuming homogeneous vehicle dynamics and negligible lateral motion, the longitudinal kinematics of the ith vehicle follow:

$$\begin{cases} \dot{p}_i\ (t) = v_i\ (t), \forall i\ \in \{0,1,\ldots,N\} \\ \dot{v}_i\ (t) = a_i\ (t), \forall i\ \in \{0,1,\ldots,N\} \\ u_i\ (t) = a_i\ (t) + T_i \dot{a}_i\,(t), \forall i\ \in \{0,1,\ldots,N\} \end{cases} \tag{1}$$

where $p_i(t) \geq 0$ m, $0 \leq v_i(t) \leq 33.3$ m/s and $-3\ \text{m/s}^2 \leq a_i(t) \leq 3\ \text{m/s}^2$ are, respectively, position, velocity, and acceleration of the ith vehicle of the platoon at time t. $u_i(t)$ is the control input, and T_i is a time constant representing the driveline delay.

The inter-vehicle spacing within the platoon is characterized by the variable time headway (VTH) as described below:

$$d_{i0} = s_{i0} + K_d \left(\alpha v_0\,(t) + \beta v_0^2\,(t)\right) \tag{2}$$

where s_{i0} is the desired standstill distance between vehicle i and platoon leader . β, α are positive relevance coefficients for the leader speed, K_d is the control parameter, and $d_{ij} = d_{i0} - d_{j0}$.

Accordingly, the relative distance $e^d_{i,i-1}$, relative velocity $e^v_{i,i-1}(t)$, and relative acceleration $e^a_{i,i-1}(t)$ during the control process can be formulated as follows:

$$\begin{cases} e^d_{i,i-1}\,(t) = p_{i-1}\ (t) - p_i\,(t) - l \\ e^v_{i,i-1}\,(t) = v_i\ (t) - v_{i-1}\,(t) \\ e^a_{i,i-1}\,(t) = a_i\ (t) - a_{i-1}\,(t) \end{cases} \tag{3}$$

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The objective of this study can be described as follows:

$$\begin{cases} \lim\limits_{t\to+\infty} \left|e^{d}_{i,0}(t)\right| = \lim\limits_{t\to+\infty} \left|e^{d}_{i,i-1}(t)\right| = 0 \\ \lim\limits_{t\to+\infty} \left|e^{v}_{i,0}(t)\right| = \lim\limits_{t\to+\infty} \left|e^{v}_{i,i-1}(t)\right| = 0 \\ \lim\limits_{t\to+\infty} \left|e^{a}_{i,0}(t)\right| = \lim\limits_{t\to+\infty} \left|e^{a}_{i,i-1}(t)\right| = 0 \end{cases} \tag{4}$$

2.2 | Bidirectional control mechanism for platoon leader

The platoon-leading vehicle must simultaneously ensure tracking safety and efficiency with the exogenous vehicle and maintain internal formation consistency. To address this complex coordination challenge, this paper proposes a bidirectional control mechanism coupling relationship between internal consistency within the platoon and the exogenous vehicle as illustrated in Figure 2.

In order to provide a description of the safety of the platoon leader during the tracking process, it is necessary to define the minimum tracking distance d_{safe} and the maximum tracking distance $d_{\max}$. Based on (Peng et al., 2023), the distance between the platoon leader and the exogenous vehicle can be estimated in the following manner:

$$\begin{cases} d_{safe}(t) = v_0(t)\cdot T_0 + (v_0(t))^2/a_{\max} + d_s \\ d_{\max}(t) = 10 + v_0(t) + 0.0825\cdot v_0(t)^2 \\ p_{e,0}(t) = p_{ex}(t) - p_0(t) - l_{ex} \end{cases} \tag{5}$$

where $a_{\max} \in [-3,3] m/s^2$ denotes acceleration limitation range, d_s is the standstill distance between adjacent vehicles. $p_{ex}(t)$ represents the position of the exogenous vehicle at time t, l_{ex} denotes for the length of the exogenous vehicle, $p_{e,0}$ represents the distance between the tail of the exogenous vehicle and the front bumper of platoon leader.

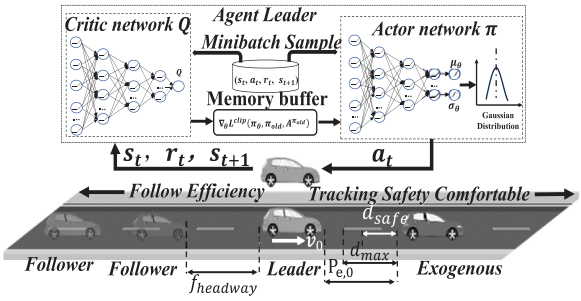


FIGURE 2 Bidirectional control mechanisms for platoon leader.

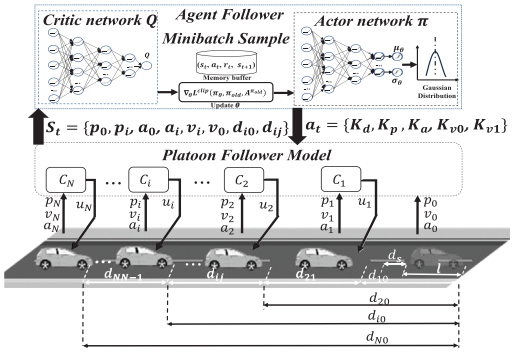


FIGURE 3 Control principle for platoon followers.

This study introduces time headway (Zhu et al., 2020) to characterize tracking efficiency behavior.

$$f_{headway}(x|\mu,\sigma) = \frac{1}{x\sigma\sqrt{2\pi}} e^{\frac{-(\ln x-\mu)^2}{2\sigma^2}} \tag{6}$$

where $x > 0$, denotes the time headway. σ and μ were set at 0.4365 and 0.4226, respectively. In this study, the optimization ranges of car following distance are set to be between the 25th and 75th percentiles, with a range of 1.15 to 1.5 s.

2.3 | MDP model for vehicle platoon followers

The consensus control model (Chen et al., 2021) accounts for communication delay and the VTH spacing strategy can be represented as

$$\begin{aligned} u_i(t) = \sum_{j=1}^{N} a_{ij} \Bigg[& K_p \begin{pmatrix} p_j(t-\tau_{ij}(t)) \\ -p_i(t) + \tau_{ij}(t) v_0(t-\tau_{ij}(t)) - d_{ij} \end{pmatrix} \\ & + K_{v0}\left(v_j(t-\tau_{ij}(t)) - v_i(t)\right)\Bigg] \\ & + a_{i0}\left[K_a\left(a_0(t-\tau_{i0}(t)) - a_i(t)\right)\right. \\ & \left. + K_{v1}\left(v_0(t-\tau_{i0}(t)) - v_i(t)\right)\right. \\ & \left. + K_p \begin{pmatrix} p_0(t-\tau_{i0}(t)) \\ +\tau_{i0}(t) v_0(t-\tau_{i0}(t)) - p_i(t) - d_{i0} \end{pmatrix}\right] \end{aligned} \tag{7}$$

where K_p, K_a, K_{v0}, K_{v1} are control parameters; $\tau_{i0}(t)$ and $\tau_{ij}(t)$ are time-varying communication delay between platoon vehicle ($j \neq i, i, j = 1, 2, \ldots, N$), and $\tau_{ij} \neq \tau_{ji}$.

As shown in Figure 3, the control instructions for the platoon followers are derived from the CCS model, while the model control parameters K_p, K_a, K_{v0}, K_{v1} are derived from the action output of the RL.

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Remark 1. This study primarily focuses on the implementation of cooperative control of the platoon through a MARL approach based on the CCS model. As such, this paper will not present a demonstration of stability analysis and convergence analysis, in pursuit of this objective, the analysis conclusions regarding the convergence have been referenced from the literature (Chen et al., 2021; Fiengo et al., 2019). The constraint ranges are: $K_d > 0, K_p > 0, K_a > 0, K_{v0} > 0, K_{v1} > 0$, and $0 \le \tau_{ij}(t) \le \tau^*, 0 \le \dot{\tau}_{ij}(t) \le h, \forall t, h \le 1$.

3 | LSTM-MAPPO CONTROL ARCHITECTURE

Section 2 delineates the control objectives for the platoon leader and followers in car-following scenarios. This section first presents the control architecture design for the platoon leader agent (Agent Leader) and the following vehicle agent (Agent Follower) utilizing the MAPPO method. Subsequently, it details the state space, action space, and cost function for both agents. The proposed control framework is illustrated in Figure 4.

3.1 | Multi-agent proximal policy optimization (PPO)

Employing an actor–critic architecture, PPO serves as a RL algorithm tailored for addressing MDP within continuous action spaces (Schulman et al., 2017). Compared to the policy gradient strategy, PPO enables mini-batch updates across multiple training iterations, addressing the issue of determining the step size. Through the importance sampling strategy, the gradient calculation $\nabla \bar{R}_\theta$ can be transformed as follows equation:

$$\nabla \bar{R}_\theta = E_{\tau \sim p_{\theta'}(\tau)} \left[\frac{p_\theta(\tau)}{p_{\theta'}(\tau)} R(\tau) \nabla log p_\theta(\tau) \right] \tag{8}$$

where $p_\theta(\tau)$ represents the probability of sequence τ occurring, $p_{\theta'}(\tau)$ is the probability of employing another NN with parameter θ' to sample data from the environment.

PPO incorporates two crucial enhancements: PPO and value function clipping. Through the comparison of the estimated state-action value functions between the new and previous policies, and leveraging the advantage function, value function clipping operations are conducted. The clipped surrogate objective function can be formulated as follows:

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, clip\left(r_t(\theta), 1-\epsilon, 1+\epsilon\right) \hat{A}_t \right) \right] \tag{9}$$

where ε is a clip fraction, and $r_t(\theta)$ is the ratio of the new policy. The generalized advantage estimator $\hat{A}(t)$ is defined as follows (Jin & Xu, 2022):

$$\hat{A}(t) \approx r(s,a) + \gamma V_\emptyset(s_{t+1}) - V_\emptyset(s_t) \tag{10}$$

MAPPO algorithm trains centrally and executes in a decentralized manner. Actor–critic networks are used to facilitate centralized training and decentralized execution. During training, exploratory behavior is used to identify the optimal strategy. The execution phase requires only forward propagation.

During the training process, the critic network utilizes global state information, which encompasses actions and observation results from all agents, to compute action-value functions. This centralized approach facilitates the evaluation of agent actions from a global perspective, aiding in decision refinement. Subsequently, the parameters of the critic network are updated by minimizing the loss.

$$loss\ (\theta^c) = E\left[Q(s, a_1, \ldots, a_N | \emptyset) - y' \right]^2 \tag{11}$$

where $y' = r_t + \gamma Q'(s', a_1', \ldots, a_N' | \emptyset')$, $s = (s_1, \ldots, s_N)$ and $\emptyset$ is the parameter of the critic network. $\emptyset'$ is the updated parameter of the evaluation network, and s' is the updated state for the target network.

To efficiently update the network parameters, the actor network iteratively updates the network parameters θ by iterating along the direction of $\nabla_\theta J(\theta)$. The global state transforms a partially observable MDP into a fully observable MDP by adopting a centralized value function. This approach enables expedited and simplified value-learning processes.

$$\nabla_\theta J(\theta) \approx E\left[\nabla_a Q(s, a_1, \ldots, a_N | \emptyset) \nabla_\theta \pi(s|\theta) \right] \tag{12}$$

3.2 | LSTM-based NN

LSTM (Pranolo et al., 2022) consists of forgetting gate f_t, input gate i_t, and output gate o_t, and these gates regulate the flow of information into and out of the cell state, which serves as the memory of the LSTM. A graphical representation of the components of the LSTM is shown in Figure 5.

When the memory cell unit processes the input x_t, certain information is deliberately discarded, while some are retained and stored. Assuming the input vector $x = (x_1, \ldots, x_{t-1}, x_t)$ and the corresponding output is $S = (S_1, \ldots, S_{t-1}, S_t)$, the gates can be defined as

$$g(x) = \sigma(W_x + b) \tag{13}$$

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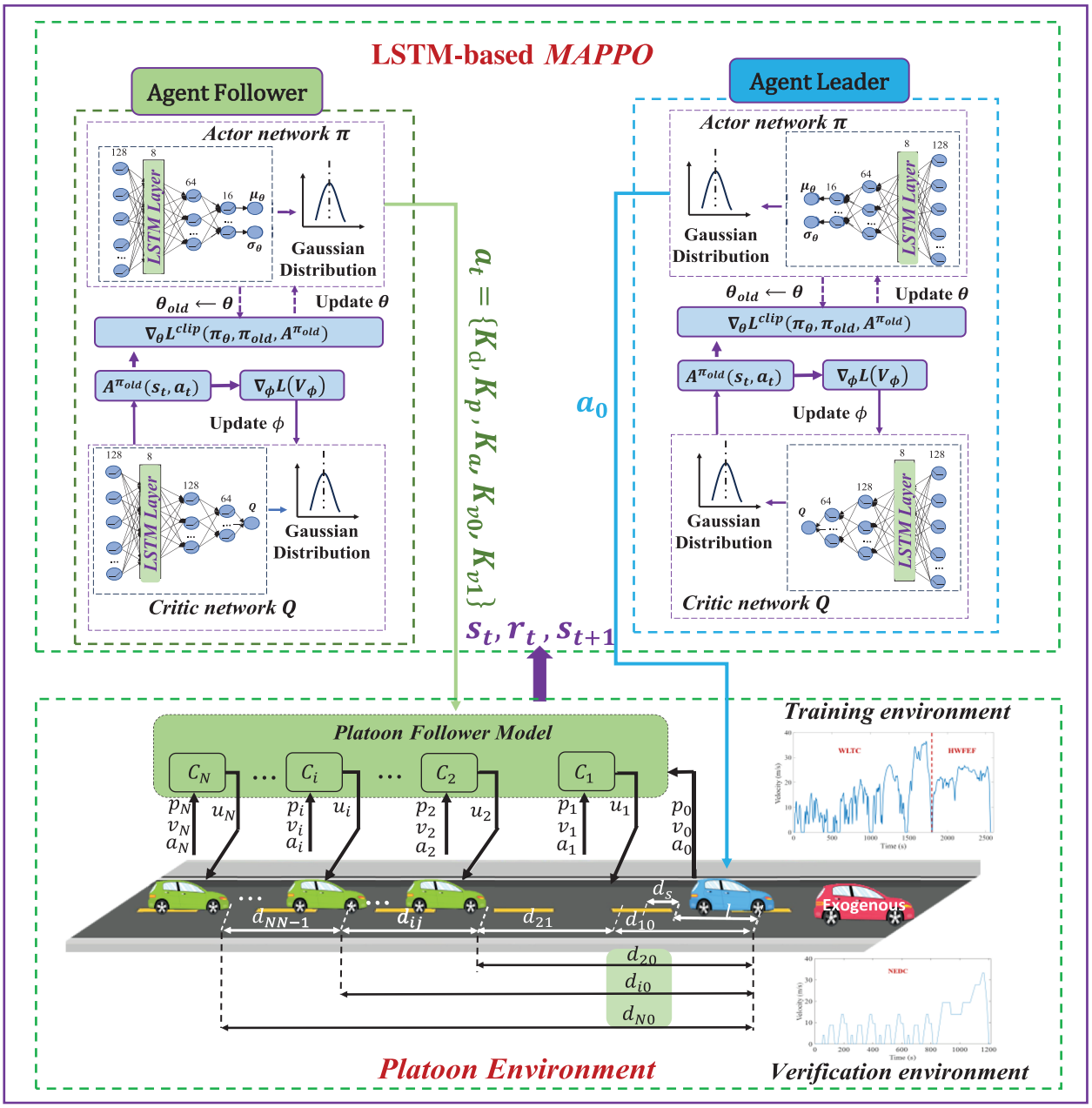


FIGURE 4 Long short-term memory (LSTM)-MAPPO control structure.

The initial stage involves the forgetting gate, which identifies the information to be discarded. Equation (14) represents the output of the forget gate.

$$f_t = \sigma\left[W_f\left(S_{t-1}, x_t\right) + b_f\right] \tag{14}$$

where σ is the sigmoid activation function, f_t denotes the forget gate with a value of $[0, 1]$, W_f denote weight matrices of forget gate, and b_f is the bias vector.

Following the forgetting gate, the vector cell state c_t is generated at the tanh level by adding the component input gate i_t to the storage cell unit for updating the information. Equation (15) corresponds to the calculation of i_t and c_t.

$$\begin{cases} i_t = \sigma\left[W_i\left(S_{t-1}, x_t\right) + b_i\right] \\ c_t = f_t.c_{t-1} + i_t.\tanh\left(W_c\left(S_{t-1}, x_t\right) + b_c\right) \end{cases} \tag{15}$$

where W_i and b_i denote for the learned parameters, W_c is the matrix weight, and b_c denotes bias vector.

The final step involves calculating the output gate o_t using the sigmoid function σ, which considers the trainable weight matrix for the output gate W_o as well as the previous output S_{t-1}, current input x_t, and bias vector b_o. Additionally, the output gate is used to multiply the tanh

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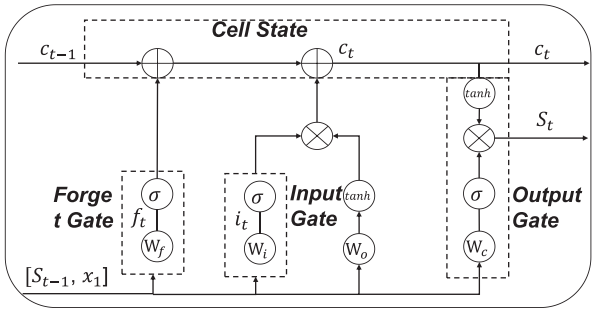


FIGURE 5 LSTM network structure.

output c_t to obtain the final output S_t.

$$\begin{cases} o_t = \sigma\left[W_o\left(S_{t-1}, x_t\right) + b_o\right] \\ S_t = o_t . \tanh\left(c_t\right) \end{cases} \tag{16}$$

3.3 | Reward, observation, and action setting

The core principle of RL revolves around the acquisition of reward $r(t)$ obtained by the agent's action $a(t)$ in response to the state $s(t)$ during its interaction with the environment. By maximizing the expectation of the cumulative discounted reward $R = E[\sum_{t=0}^{T} \gamma^t r_t]$, where T represents the total number of time steps and γ denotes the discount factor, the agent learns an optimal strategy π. Optimal control is achieved by minimizing the following cost function $K_i(t)$ in this paper:

$$r_i\ (t) = K_i\ (t) \tag{17}$$

where the subscript i denotes different agents, with number 1 representing the agent leader and number 2 representing the agent follower.

3.3.1 | Reward for agent leader

The agent leader aims to maintain a safe distance from the lead vehicle while ensuring adequate tracking performance, smooth acceleration, and efficient following behavior for the second platoon follower. Agent leader attains these objectives by maximizing the following cost function:

$$K_1\left(t\right) = \omega_1\, R_f\left(t\right) + \omega_2 R_c\left(t\right) + \omega_3 R_e\left(t\right) \tag{18}$$

where ω_1, ω_2, and ω_3 represent weights of different objectives, $R_f(t)$ is the forward safety cost, $R_c(t)$ represents for the comfortable cost, and $R_e(t)$ denotes for the backward efficiency cost. During the car-following process, penalties should be applied when vehicles exceed the maximum or minimum constraints. Therefore, the safety rewards can be listed as follows:

$$R_f = \begin{cases} p_{e,0} - d_{\max},\ if\ p_{e,0} > d_{\max} \\ -15,\ if\ d_{safe} < p_{e,0} \le d_{\max} \\ v_0,\ if\ 0 < p_{e,0} \le d_{safe} \\ v_{\max},\ if\ p_{e,0} \le 0 \end{cases} \tag{19}$$

The principle of punishment dictates that larger deviations incur greater penalties. When $p_{e,0} > d_{\max}$, it signifies inadequate following performance, resulting in a corresponding penalty. Conversely, $p_{e,0} < 0$ indicates a collision, warranting a substantial penalty equivalent to the maximum velocity $v_{\max}$.When $p_{e,0} \le d_{safe}$, the penalty value is determined based on the platoon leader's velocity v_0, with a smaller penalty assigned when the vehicle's velocity is lower, and vice versa. To incentivize optimal decision-making, a reward value of -15 is assigned to facilitate the acquisition of larger rewards.

Passenger comfort is affected by the rate of change of acceleration. To ensure that the proposed control method maintains comfort, a comfort cost function based on the rate of change of acceleration is constructed as follows:

$$R_c = \frac{{jerk_0}^2}{3600} \tag{20}$$

where $jerk_0 = (a_0(t) - a_0(t-1))/\Delta t$ denotes for the change rate of acceleration. For the sample interval of the data, Δt is set to 0.1 s, and the acceleration was bounded between $[-3\ \mathrm{m/s^2}$ and $3\ \mathrm{m/s^2}]$ in Section 2. Therefore, after squaring the jerk range will be [0 3600], this paper made a normalization setting.

The headway time between the platoon leader and the follower vehicles was considered to evaluate the backward efficiency cost function.

$$R_e = \begin{cases} \sum_{i=1}^{N} f_{headway}\left(\frac{e_{i,0}^d(t)}{v_i(t)}\right), v_i\left(t\right) > 0 \\ 0,\ else \end{cases} \tag{21}$$

3.3.2 | Reward for agent follower

Platoon followers are treated as a cohesive unit, analogous to a "small train." Therefore, the relative position $e_{i,0}^d$ and relative speed $e_{i,0}^v$ of the followers with respect to the lead vehicle are considered as trade-off factors, along with follower comfort. Consequently, the cost function can be

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constructed as follows:

$$K_2(t) = \sum_{i=1}^{N} \int_0^{\hat{t}} \omega_4 \left| e_{i,0}^{d}(t) \right| dt + \int_0^{\hat{t}} \sum_{i=1}^{N} \int_0^{\hat{t}} \omega_5 \left| e_{i,0}^{v}(t) \right| dt + \sum_{i=1}^{N} \int_0^{\hat{t}} \omega_6 \frac{jerk_i^{\,2}}{3600} dt \tag{22}$$

where ω_4, ω_5 are weight factors for distance and velocity error, and ω_6 denotes the following comfortable coefficient. $jerk_i$ denotes for the change rate of ith vehicle acceleration.

3.3.3 | State space

Based on the reward function settings and control objectives outlined above, the proposed control architecture incorporates distance deviation, speed error, velocity, and acceleration both within the platoon and between the platoon and the exogenous vehicle. The state space is defined as follows:

$$\begin{cases} s = \{s_{leader}, s_{follower}\} \\ s_{leader} = \left\{ v_0, a_0, v_{ex}, a_{ex}, e_{1,0}^{d}, p_{e,0}, e_{1,0}^{v} \right\} \\ s_{follower} = \left\{ e_{i,0}^{d}, e_{i,0}^{v}, e_{i,0}^{a}, e_{i,i-1}^{d}, e_{i,i-1}^{v}, e_{i,i-1}^{a}, \right\} \end{cases} \tag{23}$$

where the observation of the agent leader is denoted by s_{leader}, while the observation of the agent follower is represented by s_{follower}. v_{ex} and a_{ex} are corresponding to the velocity and acceleration of the exogenous vehicle, respectively.

3.3.4 | Action space

The cooperation between the two agents is designed to achieve the goal of platoon control. The agent leader's acceleration a_0 is selected to control the platoon leader. For the agent follower, the selected actions include K_d, K_p, K_a, K_{v0}, and K_{v1}, ensuring coordination within the platoon, and the combined action space such as

$$\begin{cases} a = \{a_{leader}, a_{follower}\} \\ a_{leader} = a_0 \\ a_{follower} = \{K_d, K_p, K_a, K_{v0}, K_{v1}\} \end{cases} \tag{24}$$

where a_{leader} and $a_{follower}$ are the action space of agent leader and agent follower, respectively.

3.3.5 | Training settings

As illustrated in Figure 6, a combined trip was used to create the training dataset, which consisted of the World Light Vehicle Test Cycle and the Highway Fuel Economy Test Cycle. Furthermore, the proposed algorithm was validated using the New European Driving Cycle (NEDC) scenario. The proposed framework parameters are detailed in Table 1, with the remaining simulation procedure parameters elaborated in Table 2.

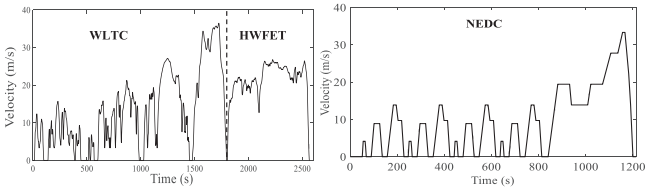


FIGURE 6 Training and validation trip: (a) combined trip for training and (b) trip for validation. HWFET, Highway Fuel Economy Test Cycle; NEDC, New European Driving Cycle; WLTC, World Light Vehicle Test Cycle.

TABLE 1 Hyperparameters used for long short-term memory (LSTM)-MAPPO.

Parameter	Agent leader	Agent follower
Actor-network size	128 128 64 16	128 128 64 16
Critic-network size	128 128 128 64	128 128 128 64
LSTM Layer	8	8
Experience horizon	1024	512
Clip factor	0.2	0.2
Entropy loss weight	0.01	0.01
Num epoch	4	4
Generalized advantage estimation (GAE) Vehicle to vehicle factor	0.95	0.95
Learning rate	[1e-4, 1e-3]	[1e-5, 1e-4]
Discount factor	0.99	0.99
Mini batch size	64	64

TABLE 2 Parameters used for simulation.

Parameter	Value
Stand still distance	$d_s = 3$ m
Vehicle parameters	$l = 5$ m, $l_{ex} = 5$ m, $T_i = 0.1$s
Spacing parameters	$\alpha = 0.0019$ s, $\beta = 0.0448$ s^2/m
Cost function parameters	$\omega_1 = -0.01, \omega_2 = -0.25, \omega_3 = 0.01$ $\omega_4 = -1, \omega_5 = -0.5, \omega_6 = -0.1$

4 | RESULTS AND DISCUSSION

To evaluate the effectiveness of the proposed control scheme, a test environment is established with one exogenous vehicle and a platoon of six vehicles. The NN

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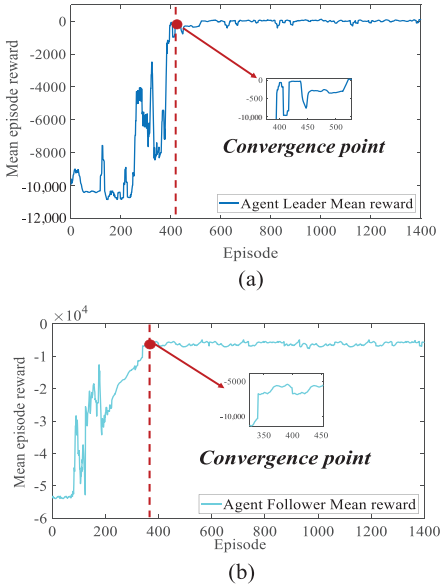


FIGURE 7 Convergence curve based on LSTM-MAPPO: (a) Mean reward of Agent leader in each training episode and (b) mean reward of Agent follower in each training episode.

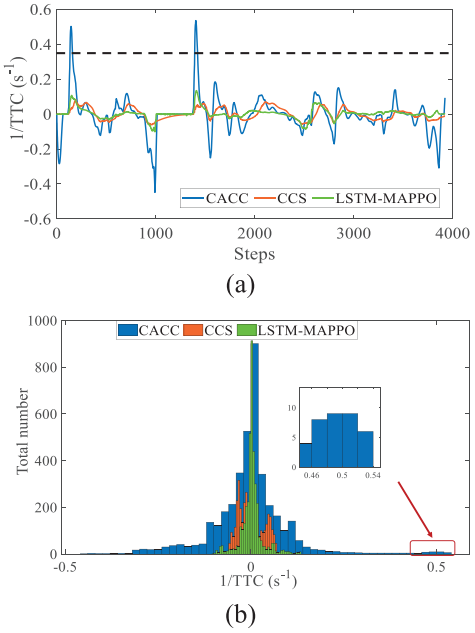


FIGURE 8 1/time-to-collision (TTC) results under different control strategies: (a) 1/TTC curve results under different control strategies and (b) 1/TTC results under different control strategies. CACC, cooperative adaptive cruise control; CCS, consensus control strategy.

parameters are trained using the combined loop illustrated in Figure 6. Subsequently, the CACC (Xiao et al., 2017), CCS (Chen et al., 2021), PID-DDPG (Yang et al., 2021), and LSTM-MAPPO are compared based on NEDC, and the method's robustness is analyzed. Specifically, the VTH spacing strategy is used to model the CACC model.

A time step of 0.1 s is configured in the simulation environment, the initial positions are $p(0) = [48, 40, 32, 24, 16, 8, 0]^T$ m, the initial velocities are consistent with the predefined trip profile, as shown in Figure 7a, and the set value is $v(0) = [0, 0, 0, 0, 0, 0, 0]^T$ m/s. Vehicle to vehicle (V2V) communication is utilized between the platoon vehicles to enable information exchange of vehicle position, speed, and acceleration. Additionally, the time delay of neighboring vehicles passing through the V2V is 5 ms.

4.1 | Training process and convergence analysis

Figure 7 presents the temporal evolution of average episodic rewards across 1400 training rounds. The learning curves demonstrate progressive improvement for both leader and follower agents, stabilizing at elevated performance levels after distinct convergence phases. Specifically, convergence is achieved by the leader agent at 400 epochs and the follower agent at 340 epochs. This stable co-evolution pattern validates the efficacy of our dual-agent MARL architecture in coordinating platoon control objectives.

4.2 | Training and testing results of platooning scenario

4.2.1 | Bidirectional control mechanisms analysis

Time-to-collision (TTC) serves as a critical safety indicator for quantifying collision risks in vehicular platoons (Cui et al., 2022). Higher 1/TTC value values correlate with increased oscillation risks during vehicle-approaching maneuvers. To ensure a clear comparison of the test results, this paper s analysis on the period of 0–390 s within the scenario. In Figure 8a, the critical TTC threshold value is represented by the black dotted line, which is set to 3 s (Yuan Zheng et al., 2023). Notably, the CACC strategy shows the most significant change, exceeding a magnitude of 0.4. Conversely, the CCS and LSTM-MAPPO methods exhibit smaller variations, with none surpassing the black dashed boundary value of 0.2. Figure 8b shows the bar result plot, which indicates that the distribution of the 1/TTC value for the LSTM-MAPPO control strategy is concentrated within [−0.1 0.1], displaying minimal fluctuations overall. This result validates LSTM-MAPPO's superior safety performance through stabilized vehicle interactions, particularly during dynamic traffic scenarios.

To analyze the impact of bidirectional control on efficiency, this paper examined its effect on the second vehicle.

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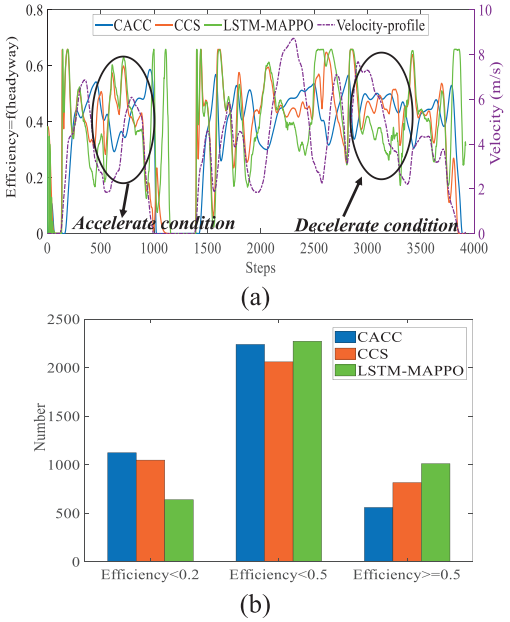


FIGURE 9 Tracking efficiency of the second follower: (a) Efficiency results curves and (b) distribution of the efficiency results. CACC, cooperative adaptive cruise control; CCS, consensus control strategy.

Figure 9a compares the efficiency curves, demonstrating that LSTM-MAPPO achieves the highest efficiency (close to 0.65), followed by CCS. CACC exhibits the lowest efficiency (around 0.3) during acceleration. During deceleration, CACC showed improved efficiency (around 0.5), compared to CCS and LSTM-MAPPO. Furthermore, Figure 9b shows the efficiency distribution, indicating that the LSTM-MAPPO approach has a wider efficiency range (total 1010 time steps exceeded 0.5), representing approximately 25.7% of the total.

4.2.2 | Platooning performance analysis

To facilitate an overall comparison, this subsection focuses on analyzing a specific time range of 70–90 s (200 steps) within the driving cycle scenario, simultaneously, three control methods, CACC, CCS, and LSTM-MAPPO, are selected for comparison.

Figure 10 illustrates that the LSTM-MAPPO strategy resulted in a greater mean relative distance (21.7 m) to the exogenous vehicle, compared to CCS (19.51 m). Furthermore, LSTM-MAPPO effectively controlled the relative distance of the followers to approximately 11.5 m, compared to CACC (25.1 m) and CCS (19.1 m). The results indicate that the platoon leader maintains a larger following distance with exogenous vehicles, compared to the CCS strategy, while simultaneously improving the overall following performance of the followers.

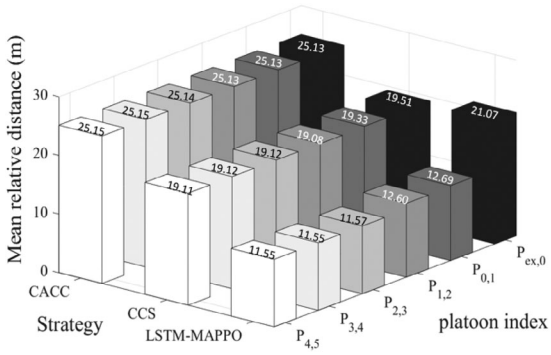


FIGURE 10 Mean relative distance of the platoon. CACC, cooperative adaptive cruise control; CCS, consensus control strategy.

Figure 11a–c shows an increased distance between the platoon leader and the external vehicle when using LSTM-MAPPO. However, at Step 200, the position of the sixth follower shows the largest outcome (1132.78 m), indicating an improvement in platoon following efficiency.

Figure 11d–f shows that the CACC-based control strategy results in larger relative speeds among platoon followers, while the proposed control strategy exhibits the smallest relative speeds. Furthermore, as depicted in Table 3, the sum of absolute speed errors for all vehicles over the entire observation window is significantly smaller for LSTM-MAPPO (57.64 m/s), compared to CCS (96.68 m/s) and CACC (206.1 m/s). These results indicate the superior speed regulation performance of the proposed strategy, compared to CACC and CCS.

Figure 11g–i illustrates the jerk results for the three control algorithms. Figure 11g shows smoother fluctuations for CACC, Figure 11h illustrates larger fluctuation amplitudes for CCS, and Figure 11i shows a higher fluctuation frequency for LSTM-MAPPO. This is attributable to the nature of the MDP, which leads to higher jerk fluctuation frequencies; however, the reward mechanism promotes better overall performance. As shown in Table 3, the sum of jerk through CACC (59.06 m/s^3) is larger than CCS (34.94 m/s^3) and LSTM-MAPPO (34.02 m/s^3). Regarding the mean squared error (MSE), LSTM-MAPPO (0.13 m/s^3) outperforms CACC (0.33 m/s^3) and CCS (0.21 m/s^3).

It has been demonstrated that disturbances in platoon spacing and velocity errors do not propagate within the platoon, a phenomenon known as string stability (Feng et al., 2019). In order to evaluate the efficacy of the algorithm, the time of 0–390 s within the simulation scenario is utilized to examine string stability. It can be observed from Figure 12 that the distance error and velocity error of the platoon leader are dampened as they propagate to the platoon downstream. This demonstrates the platoon maintaining string stability.

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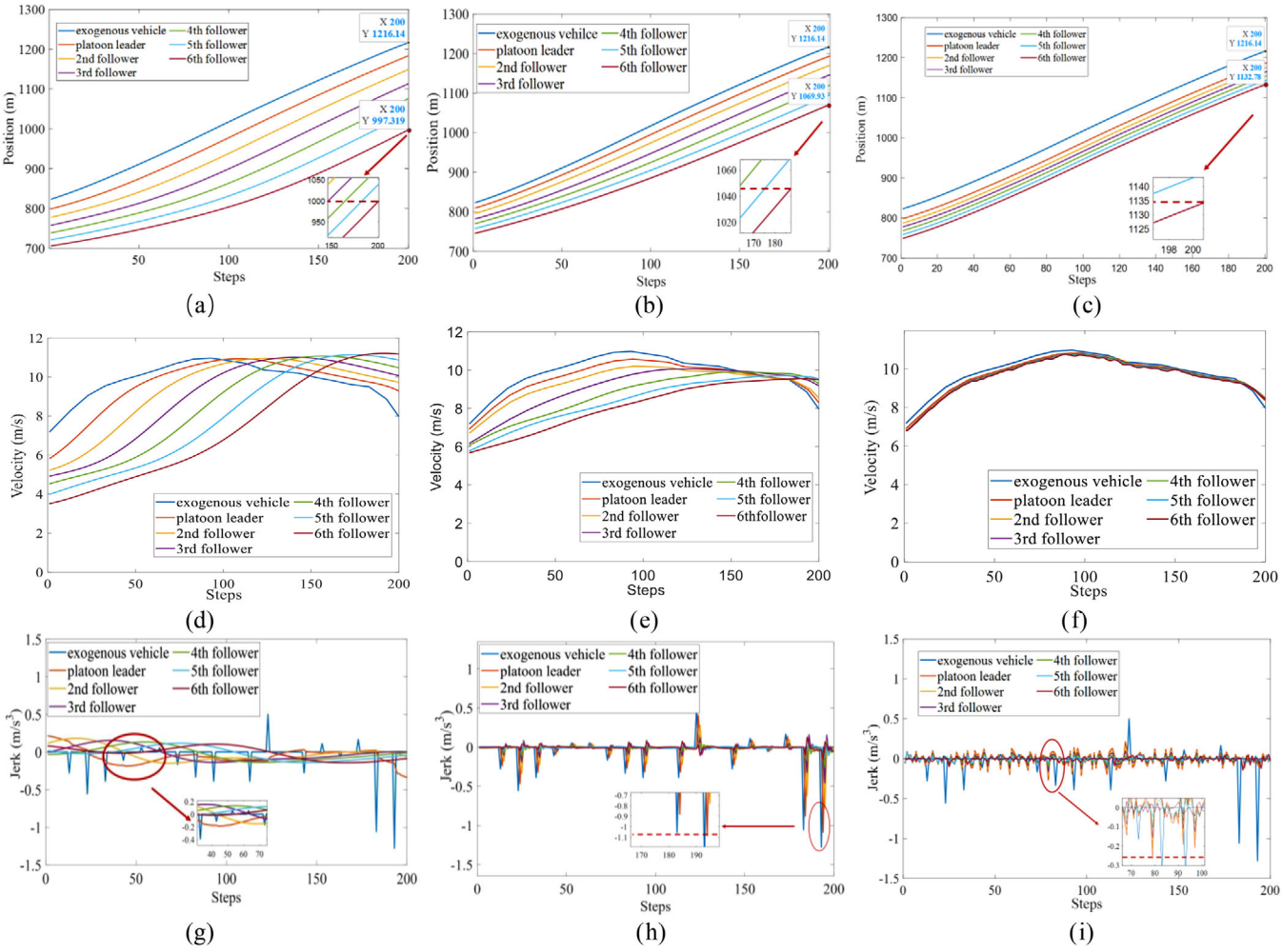


FIGURE 11 Compare results under different strategies: (a) cooperative adaptive cruise control (CACC)-based trajectory profile, (b) consensus control strategy (CCS)-based trajectory profile, (c) LSTM-MAPPO-based trajectory profile, (d) CACC-based velocity profile, (e) CCS-based velocity profile, (f) LSTM-MAPPO-based velocity profile, (g) CACC-based jerk profile, (h) CCS-based jerk profile, and (i) LSTM-MAPPO-based jerk profile.

TABLE 3 Simulation results comparison.

Item	Cooperative adaptive cruise control (CACC)	Consensus control strategy (CCS)	Proposed
$\sum_{i=1}^{2}\sum_{t=1}^{200} r_i(t)$	−137.06	−73.87	−41.61
$\sum_{i=1}^{6}\sum_{t=1}^{200} \lvert\Delta v_{i,i-1}(t)\rvert$	206.1	96.68	57.64
$\sum_{i=0}^{5}\sum_{t=1}^{200} \lvert jerk_i(t)\rvert$	59.06	34.94	34.02
$\sqrt{\sum_{i=0}^{5}\sum_{t=1}^{200}\left(jerk_i(t)-\overline{jerk}_i\right)^2/N}$	0.33	0.21	0.13

4.3 | Validation and discussion

This subsection validates the effectiveness of the proposed method based on the typical driving cycle scenario NEDC. It compares it with the MPC, ISMC, PID-DDPG, CCS, and CACC* (based on constant time headway [CTH], set 0.6 s) approach.

4.3.1 | Validation in NEDC scenarios

Figure 13 presents the validation results in the NEDC scenario. From Figure 13a, it can be observed that the vehicle platoon maintains a safe operational distance from the exogenous vehicle, with no collisions occurring. On the basis of the proposed method, the spacing within

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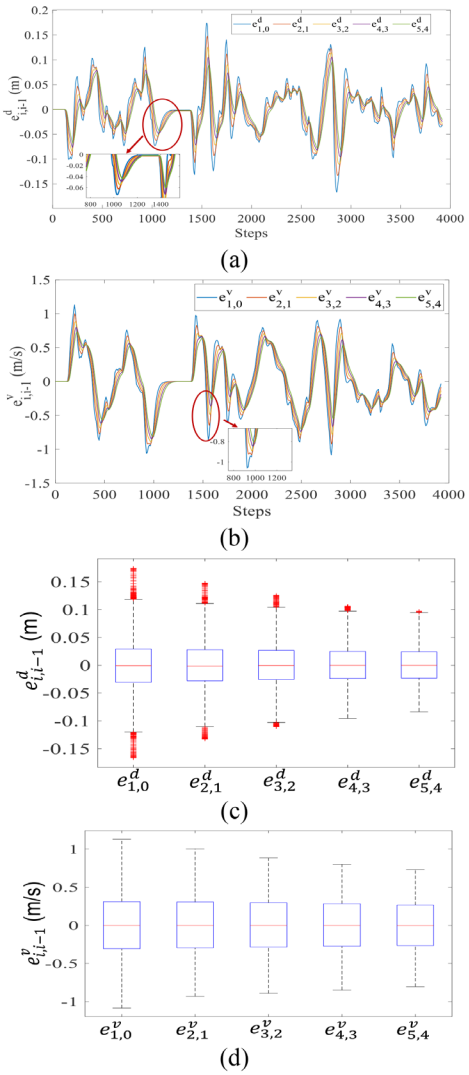


FIGURE 12 Results of distance error and velocity error based on LSTM-MAPPO: (a) distance error profile, (b) velocity error profile, (c) box of distance error, (d) box of velocity error.

the platoon resembles that of a "small train." Figure 13b demonstrates that the speed consistency between the vehicles in the platoon is well-maintained, with minimal variations in relative speed during acceleration and deceleration processes. More detailed validation results can be found in Table 4. While the inter-vehicle distance between the platoon leader and exogenous vehicle remains comparatively large, the proposed methodology yields significant improvements in average platoon length, relative speed error, relative acceleration, and average acceleration variance.

4.3.2 | Ablation and comparison

The experimental validation of the bidirectional coordination mechanism and LSTM-enhanced architecture is

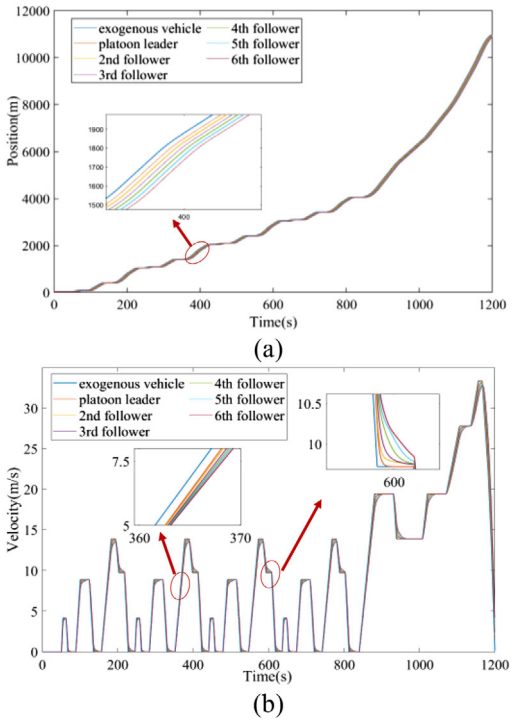


FIGURE 13 Validation curves for New European Driving Cycle (NEDC) scenarios based on LSTM-MAPPO: (a) trajectory profile and (b) velocity profile.

TABLE 4 Validation statistics for New European Driving Cycle scenarios.

Measurement	CCS	Proposed
$\sum_{t=1}^{12100} p_{e,0}(t)$	357,397	371,250
$\sum_{t=1}^{12100} e_{5,0}^{d}(t)/N$	95.94	77.58
$\sum_{i=1}^{6}\sum_{t=1}^{12100} \lvert e_{i,i-1}^{d}(t)\rvert$	12,676.26	9707.13
$\sum_{i=1}^{6}\sum_{t=1}^{12100} \lvert e_{i,i-1}^{v}(t)\rvert$	7225.54	5099.67
$\sum_{i=0}^{5}\sum_{t=1}^{12100} \lvert jerk_i(t)\rvert$	2468.78	1696.1
$\sqrt{\sum_{i=0}^{5}\sum_{t=1}^{12100}\left(jerk_i(t)-\overline{jerk_i}\right)^2/N}$	0.29	0.19

systematically demonstrated in Table 5. When compared to the proposed experiments, it is evident that there are significant improvements in the other evaluated metrics besides the increase in following distance with exogenous vehicles. The standard deviation of jerk shows a significant improvement, indicating that the bidirectional mechanism effectively optimizes anomalous acceleration fluctuations within the platoon. Furthermore, the table showcases the results obtained from prior research utilizing the PID-DDPG method. The suggested control approach has resulted in substantial improvements in managing platoon length, speed error, and spacing error. Regarding the total distance error, total velocity error, MSE of the jerk, the LSTM-MAPPO-based approach outperforms the CCS by 8.3%, 2.46%, and 12.5%. Based on the results, it is evident

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TABLE 5 Comparison of results for different control strategies.

Measurement	CACC*	Model predictive control (MPC)	Integral sliding-mode control (ISMC)	PID-deep deterministic policy gradient (DDPG)	MAPPO (with bidirectional)	LSTM-MAPPO (no bidirectional)	Proposed
$\sum_{t=1}^{12100} p_{e,0}(t)$	337,012	342,503	349,971	377,976	387,651	344,115	371,250
$\sum_{t=1}^{12100} e_{5,0}^{d}(t)/N$	186.32	95.71	101.32	93.94	83.32	101.41	77.58
$\sum_{i=1}^{6}\sum_{t=1}^{12100} \lvert e_{i,i-1}^{d}(t)\rvert$	9491.17	10,302.12	10,864.16	11,002.91	9996.81	11,401.38	9707.13
$\sum_{i=1}^{6}\sum_{t=1}^{12100} \lvert e_{i,i-1}^{v}(t)\rvert$	9293.66	6719.33	6618.43	5719.33	5817.98	5907.46	5099.67
$\sum_{i=0}^{5}\sum_{t=1}^{12100} \lvert jerk_i(t)\rvert$	2841.13	2064.21	2736.95	1904.21	1901.94	1871.45	1696.1
$\sqrt{\sum_{i=0}^{5}\sum_{t=1}^{12100}\left(jerk_i(t)-\overline{jerk_i}\right)^2/}$	0.37	0.24	0.33	0.25	0.21	0.24	0.19

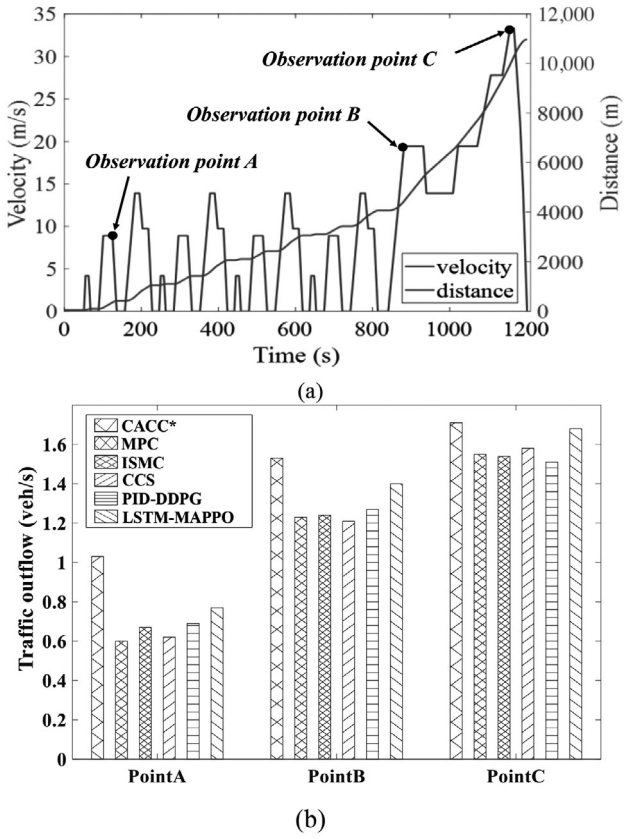


FIGURE 14 Traffic outflow comparison: (a) observation points based on the NEDC scenario and (b) traffic outflow under different strategies. DDPG, deep deterministic policy gradient; ISMC, integral sliding-mode control; MPC, model predictive control.

that the proposed method exhibits a larger total spacing with the exogenous vehicle. By comparing with MPC, ISMC, and PID-DDPG methods show similar results in terms of following efficiency, while the MSE of the jerk of the ISMC method is relatively large, but in general, the proposed method overall shows better results.

4.3.3 | outflow

The traffic outflow (Q) can be calculated as the total number of vehicles (N_{total}) divided by the time interval ($t_1 \sim t_n$) between the passing vehicle and the identical downstream measurement position on the roadway as expressed by the following formula (Duret et al., 2020):

$$Q = \frac{N_{total}}{t_n - t_1} \tag{25}$$

To ensure comprehensive observation across various driving conditions, representative points A (125 s), B (880 s), and C (1156 s) were selected from the NEDC as shown in Figure 14a. These points capture a diverse range of speeds,

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TABLE 6 Time-exposed time-to-collision (TTC; TET) and time-integrated TTC (TIT) results comparison.

Measurement	Strategy	TTC^* = 1	TTC^* = 2	TTC^* = 3
TET	CACC*	1.17	4.04	9.17
	Model predictive control (MPC)	0.00	0.00	0.00
	Integral sliding-mode control (ISMC)	0.00	0.00	0.00
	CCS	0.00	0.00	0.00
	PID-DDPG	0.00	0.00	0.00
	LSTM-MAPPO	0.00	0.00	0.00
TIT	CACC*	0.81	1.16	2.54
	MPC	0.00	0.00	0.00
	ISMC	0.00	0.00	0.00
	CCS	0.00	0.00	0.00
	PID-DDPG	0.00	0.00	0.00
	LSTM-MAPPO	0.00	0.00	0.00

enabling a thorough analysis of the system's performance under low, medium, and high-speed conditions.

As demonstrated in Figure 14b, the traffic outflow is observed under various control strategies. Overall, CACC is observed to have the highest outflow due to the fixed headway strategy. In comparison to the other strategies, LSTM-MAPPO is observed to have the highest outflow, followed by PID-DDPG, ISMC, and MPC. At high speeds, MPC, CCS, and ISMC produce similar outflows of 1.55, 1.54, and 1.58 veh/s, respectively. LSTM-MAPPO has the highest flow outflow of all the strategies at 1.68 veh/s. Further analysis indicates that the LSTM-MAPPO-based control method enhances the flow outflow by 30.4%, 28.6%, and 7.5% in low-, medium-, and high-speed scenarios, respectively, when compared to CCS.

4.3.4 | Safety

Two derived TTC metrics—time-exposed TTC (TET) and time-integrated TTC (TIT)—were employed to quantify traffic conflict risks, following the methodology established by Yao et al. (2023).

Table 6 compares vehicle platoons under various TTC^* safety values; the CTH strategy exhibits higher TIT and TET values, suggesting potentially compromised safety performance. In contrast, the LSTM-MAPPO approach inherits the control benefits of CCS while maintaining good safety performance, whereas CACC* exhibits relatively lower safety performance. This validation result aligns with Figure 8a, where 1/TTC values for both LSTM-MAPPO and CCS in the training scenario are distributed within [−0.1, 0.1], as shown in Figure 8b.

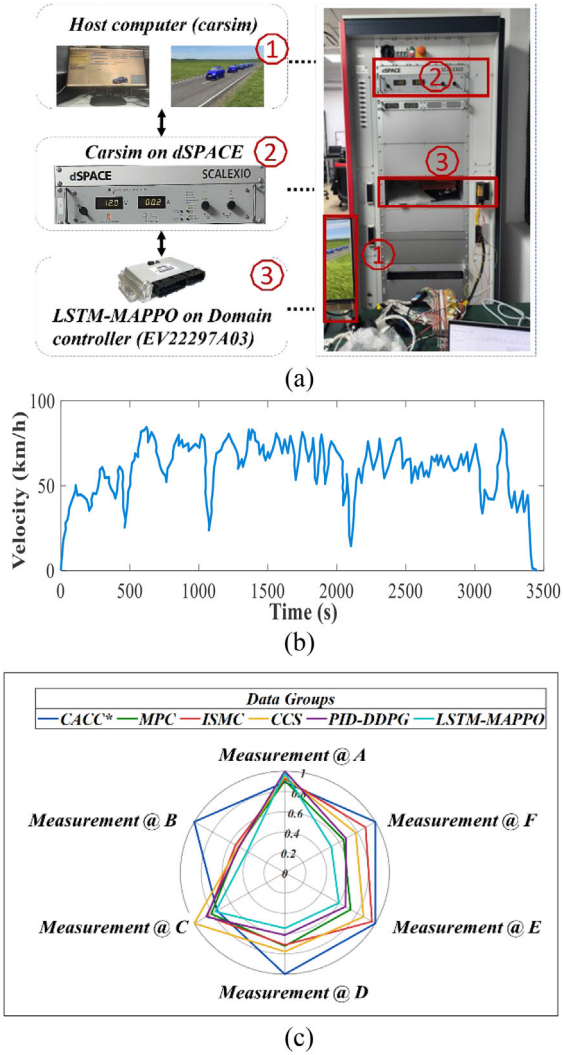


FIGURE 15 Hardware-in-the-loop test setup: (a) test environment setup, (b) test road profile, and (c) test results. DDPG, deep deterministic policy gradient; ISMC, integral sliding-mode control; MPC, model predictive control.

4.4 | Analysis of hardware-in-the-loop (HIL) test results

HIL testing plays a critical role in validating the performance and reliability of proposed strategies under real-world application scenarios. The constructed HIL system, as illustrated in Figure 15a, comprises the following components: CarSim software for vehicle kinematics modeling, MATLAB/Simulink software for control algorithm development, a domain controller EV22297A03 (Infineon TC297), a dSPACE real-time system for executing high-fidelity simulations, and a host computer for system

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monitoring and data acquisition. This integrated platform ensures seamless communication and synchronization across subsystems, enabling rigorous validation of control strategies under dynamically emulated operational conditions.

As illustrated in Figure 15b, the road condition data utilized for HIL testing were sourced from real-world road conditions (Han et al., 2022). A systematic parameter-by-parameter comparison was conducted following the experimental protocol defined in Table 5, utilizing six evaluation metrics (denoted as Measure@A to Measure@F) for comprehensive performance assessment. Figure 15c illustrates the statistics of the normalized results, where the proposed method improves 11.7%, 16.4%, 24.1%, and 29.3% in terms of velocity error, distance error, jerk, and jerk MSE, respectively, as compared to the conventional benchmark method CCS. This empirical evidence substantiates the enhanced robustness and adaptability of the LSTM-MAPPO framework in managing complex dynamic interactions under practical HIL testing conditions, thereby providing a foundational basis for further industrial applications.

5 | CONCLUSION AND FUTURE WORK

This paper investigated the integration of MARL and CCS to address platoon control challenges while preserving the coupling relationship between internal coordination within the platoon and the exogenous vehicle. Multi-agent Markov game theory created a joint target value function estimated by LSTM to capture time-relatedness in platoon control. The platoon's performance improved, convergence stability improved, and traffic flow efficiency enhanced through training in a mixed driving scenario. The primary conclusions are summarized as follows:

1. The LSTM-MAPPO algorithm achieves a 19.13% reduction in average platoon length compared to CCS, attributed to the bidirectional control mechanism.
2. The incorporation of the LSTM network has resulted in improved traffic outflow performance compared to the original CCS method. This improvement is evident across various scenarios, with a 20.1% increase in optimization effect observed in low-speed scenarios. The proposed control strategy has demonstrated an 8.3% reduction in spacing error control in high-speed scenarios.

While the LSTM-MAPPO framework enhances platoon coordination, three limitations persist: first, homogeneous assumptions misrepresent mixed fleets. Second, V2X uncertainties degrade decision synchronization. Third, labeled data dependency limits scenario generalization. Future work will explore the following: neural dynamic classification for heterogeneous interaction pattern recognition through dynamic weighting; dynamic ensemble learning for robust multi-sensor fusion under communication disturbances; finite element machines for sparse state representation; and self-supervised strategies for latent feature extraction. These aim to build universal cooperative control while retaining computational efficiency.

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